SIMATS ENGINEERING
Slot: A, Course Code: ITA0513, Course Name: Computer Vision

Duration: 60 mins **Marks: 20**
Assessment Tool 1: Short Answer Test (High) **Assessment Tool Weightage: 40%**

COURSE OUTCOME COVERED

CO1: *Apply the fundamental concepts, image formation models, and processing techniques for analyzing Computer Vision systems.* (BTL 3)

Q. No.	Questions
1	A factory uses cameras to inspect products moving on a conveyor belt. Explain how Computer Vision can identify defective products automatically.
2	A facial recognition system is affected by changes in pose and illumination. Explain which Computer Vision features can help improve recognition performance.
3	A traffic monitoring system needs to detect vehicles, recognize traffic signs, and estimate distances. Explain how different levels of Computer Vision are involved.
4	A scanned document contains blurred characters and unwanted noise. Explain how digital image fundamentals can be used to improve the document before further processing.
5	A smartphone camera produces different image quality under indoor and outdoor conditions. Explain the factors involved in image sensing and acquisition.
6	A surveillance camera captures a moving person as a blurred object. Identify possible causes related to image acquisition and suggest suitable improvements.
7	A digital image contains very few pixels and important object details are missing. Explain how the sampling process affects the image quality.
8	An image contains smooth intensity variations, but after digitization it shows noticeable intensity steps. Explain how quantization causes this effect.
9	A remote sensing system must capture very small objects on the Earth's surface. Explain how the choice of sampling resolution affects the ability to detect these objects.
10	A grayscale image is stored using 2 bits per pixel instead of 8 bits per pixel. Explain how this change affects the number of intensity levels and image appearance.

Code of Conduct:

*I **Devadharshini M (192572149)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student
Devadharshini M
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RUBRICS FOR CALCULATION

Criteria	Weigh tage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand

ITA0513 - Computer Vision

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CO1 - AT1 - Short Answer Test

1. Defective Product Detection:

⇒ Computer Vision uses a camera to capture images of products on the conveyor belt. ⇒ Image procesing techniques such as segmentation, edge detection, and feature extraction.

2. Facial recognition:

⇒ Facial recognition can be improved using features that are less affected by pose and lightning such as edges, local features and facial landmarks.

⇒ Techniques like image normalization, histogram can affect the lighting

2. Traffic monitoring using computer vision:

⇒ Cameras capture road image/videos.

⇒ Object detection identifies vehicles

⇒ It tracks to identify and follow vehicle.

⇒ It can calculate vehicle speed, distance and traffic flow.

Scanned document:

⇒ Filtering removes unwanted noise, while sharpening improves blurred characters.

⇒ Thresholding separates the text from the background for better recognition

5. Smartphone image quality:

⇒ Image quality depends on lighting, exposure, ISO and shutter speed.

⇒ Indoor low light may cause noise, while strong outdoor light can cause overexposure.

6. Moving person appears blurred:

⇒ Motion blur occurs when they are moving while the camera shutter is open.

⇒ It can be reduced using a faster shutter speed and better lighting.

7. Very few pixels:

⇒ Low sampling produces fewer pixels and causes loss of important image details.

⇒ Higher sampling gives better resolution and clear objects.

8. Intensity steps after digitization:

⇒ This occurs due to quantization, where continuous intensity values are converted into limited levels.

⇒ More intensity levels reduce the visible steps and improve image quality.

9. Remote sensing:

⇒ Spatial resolution determines the amount of detail visible in an image.

⇒ Higher spatial resolution uses smaller pixels, making small objects easier to detect.

10. 2 bits Vs 8 bits:

⇒ 2 bits provide $2^2 = 4$ intensity levels, while 8 bits provide $2^8 = 256$ levels.

⇒ ∴ 2-bit images have less intensity variation and may look less smooth.